MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (FOUR PLIES, SPIRAL-WRAPPED OR TWO PLIES, SPIRAL-WRAPPED AND ONE PLY, BRAIDED HIGH PRESSURE SIZES -6 THROUGH -16)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by four plies of spiral wrapped or two plies spiral wrapped and one ply braided wire in sizes -6 through -16 for high pressure hydraulic applications.

1.2 Classification. End fittings for hoses covered by this specification shall be sizes -6 through -16 on the hose end as applicable and shall be of the following types:

Type A - SAE, 37-degree tube swivel to hose, straight, tube sizes 3/8 inch to 1-1/4 inches (see figure 1).
Type B - Gasket seal union, male half to hose, straight, tube sizes 1/4 to 1 inch (see figure 2).
Type BL - Gasket seal union, male half to hose, 90-degree elbow, tube sizes 1/4 to 3/4 inch (see figure 3).
Type C - O-ring seal union, male half to hose, straight, tube sizes 1/4 to 1 inch (see figure 4).
Type CL - O-ring seal union male half to hose, 90-degree elbow, tube sizes 1/4 to 3/4 inch (see figure 5).
Type D - 90-degree elbow, hose to hose, hose dash size -6 through -16 (see figure 6).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-L-17731 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. The rubber compounds used shall be resistant to petroleum base oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

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3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 6.

3.2.2.1 Socket. Socket for fitting assemblies shall be one-piece design and shall not exceed the envelope dimensions shown on figures 1 through 6.

3.3 Reinforcement. Reinforcement shall consist of four plies spiral wrapped wire applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dimensions and pressures.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)						Min bend radius (inches)	Approx wt/ft (lbs/ft)
			Id tube		Od cover		Od reinf			
	Working (max)	Burst (min)	min	max	min	max	min	max		
-6	4,500	18,000	0.367	0.398	0.812	0.875	0.665	0.711	5	0.46
-8	4,000	16,000	.485	.531	.938	1.000	.766	.828	7	.64
-12	3,000	12,000	.735	.781	1.203	1.265	1.047	1.109	9.5	.82
-16	2,500	10,000	.985	1.040	1.516	1.609	1.360	1.422	12	1.16

3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

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4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-6	4
-16	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

TABLE III. Number of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-6	8
-16	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-6	2	2	-6	4	4
-16	3	2	-16	5	5

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-6	2	1	1	-6	4	2	2
-16	3	1	1	-16	5	3	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers								
	Size (dash no.) -6				Size (dash no.) -16				
Test sequence	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X			
Impulse test (see 4.6.5)1/				X				X	X
Burst test (see 4.6.6)1/	X				X				
Cold temperature flexing (see 4.6.8)1/			X				X		

1/ See MIL-H-24135.

4.3 The test fluid shall be petroleum base oil in accordance with MIL-L-17331 or MIL-L-2104 for the aging, hydraulic fluid circulation, and impulse tests.

4.4 The peak pressure for the impulse test shall be 150 percent of maximum working pressure in accordance with table I.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

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6. NOTES

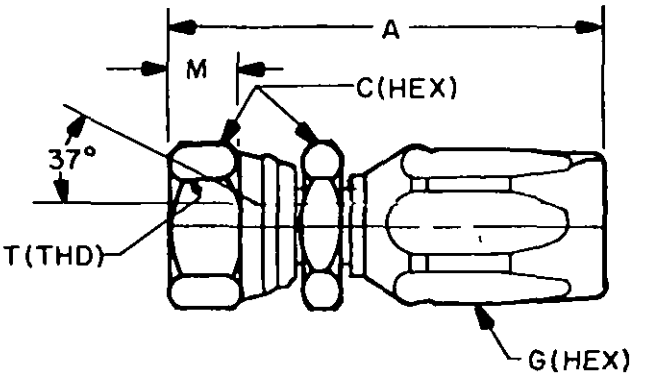
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems where pressures do not exceed the rated working pressure and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

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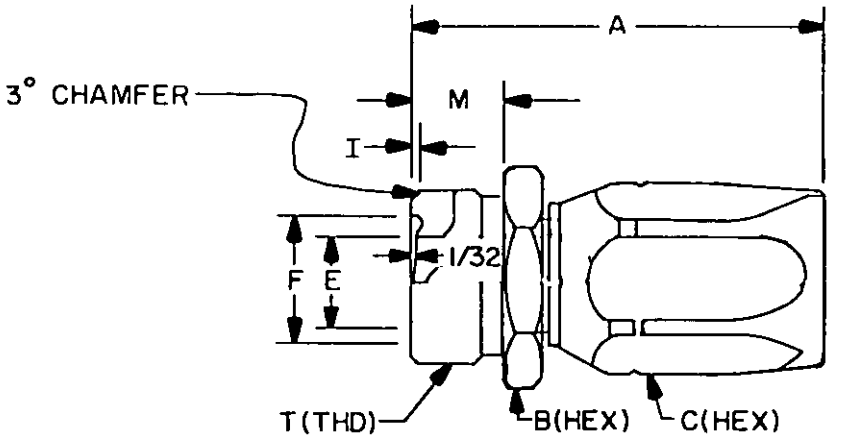
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	C (see note 2)	G (see note 2)	M (ref)
3/8	-6	9/16-18 UNF	3.00	0.688	1.25	0.40
1/2	-6	3/4-16 UNF	3.25	.875	1.25	.44
1/2	-8	3/4-16 UNF	3.50	.875	1.50	.44
5/8	-8	7/8-14 UNF	3.50	1.00	1.50	.52
3/4	-8	1-1/16-12 UNF	3.50	1.25	1.50	.57
3/4	-12	1-1/16-12 UN	4.00	1.25	1.75	.57
7/8	-12	1-3/15-12 UN	4.00	1.375	1.75	.58
1	-12	1-5/16-12 UN	4.00	1.50	1.75	.60
1	-16	1-5/16-12 UN	4.50	1.50	2.25	.60
1-1/4	-16	1-5/8-12 UN	4.75	2.00	2.25	.64

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for swivel nut, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn, or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 1. Type A, SAE 37-degree tube swivel to hose, straight.

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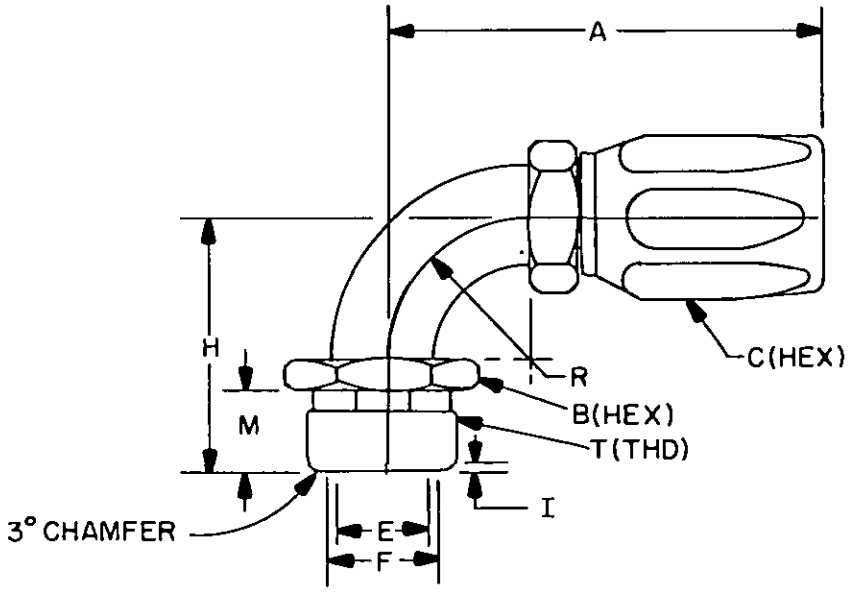
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (see note 2)	C (see note 2)	E ± 0.005	F ± 0.005	I (ref)	M (ref)
1/4	-6	1-1/8-12 UNF	3.25	1.25	1.25	0.406	0.641	1/16	0.88
3/8	-6	1-1/4-12 UNF	3.25	1.375	1.25	.547	.797	1/16	.88
3/8	-8	1-1/4-12 UNF	3.50	1.375	1.50	.547	.797	1/16	.88
1/2	-12	1-1/2-12 UNF	4.00	1.625	1.75	.672	1.000	1/16	.94
3/4	-16	1-3/4-12 UN	4.50	1.875	2.25	.875	1.188	5/64	1.00
1	-16	2-1/8-12 UN	4.75	2.25	2.25	1.094	1.375	5/64	1.12

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.
4. This fitting is for replacement purposes only and is not to be used in new construction.

FIGURE 2. Type B, gasket seal union, male half to hose, straight.

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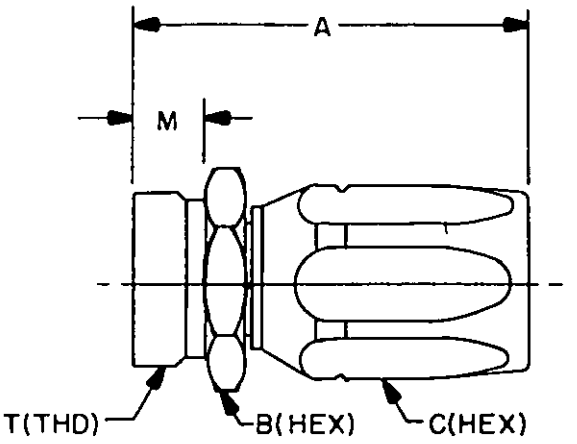
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (see note 2)	C (see note 2)	H (max)	E ± 0.005	F ± 0.005	I (ref)	M (ref)	R rad (min)
1/4	-6	1-1/8-12 UNF	3.55	1.25	1.25	2.25	0.406	0.641	1/16	0.88	0.50
3/8	-8	1-1/4-12 UNF	3.81	1.375	1.50	2.50	.547	.797	1/16	.88	.75
1/2	-12	1-1/2-12 UNF	4.75	1.625	1.75	3.25	.672	1.000	1/16	.94	1.25
3/4	-16	1-3/4-12 UN	5.75	1.875	2.25	3.75	.875	1.188	5/64	1.00	1.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adaptor hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. The 90-degree tube elbow for fitting sizes 1/4 through 3/4 inch shall conform to MIL-T-1368, class A.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.
5. This fitting is for replacement purposes only and is not to be used in new construction.

FIGURE 3. Type BL, gasket seal union, male half to hose, 90-degree elbow.

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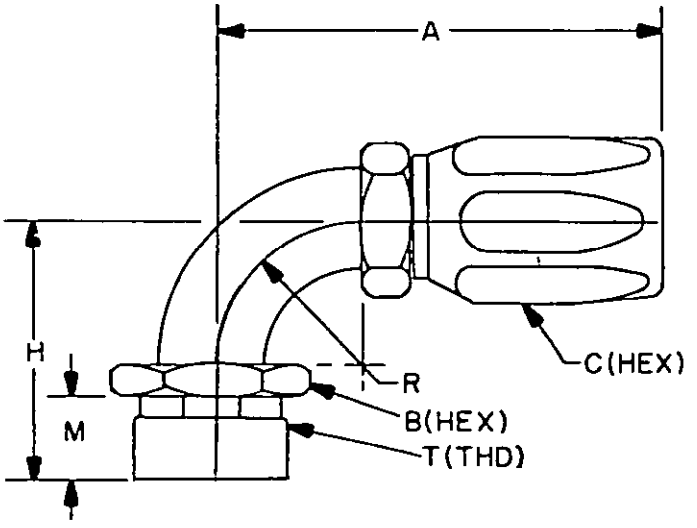
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Tube size (nom)	Hose end size	T thread CL-3A	A (max)	B (see note 2)	C (see note 2)	M (ref)
1/4	-6	1-3/16-12 UN	3.25	1.25	1.25	0.62
3/8	-6	1-3/8-12 UNF	3.25	1.44	1.25	.62
3/8	-8	1-3/8-12 UNF	3.50	1.44	1.50	.62
1/2	-12	1-3/4-12 UN	4.00	1.81	1.75	.62
3/4	-16	2-12 UN	4.50	2.12	2.25	.62
1	-16	2-5/16-12 NS	4.75	2.38	2.25	.75

NOTES:
1. Dimensions are in inches.
2. Fitting assembly materials for hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 4. Type C, O-ring seal union, male half to hose, straight.

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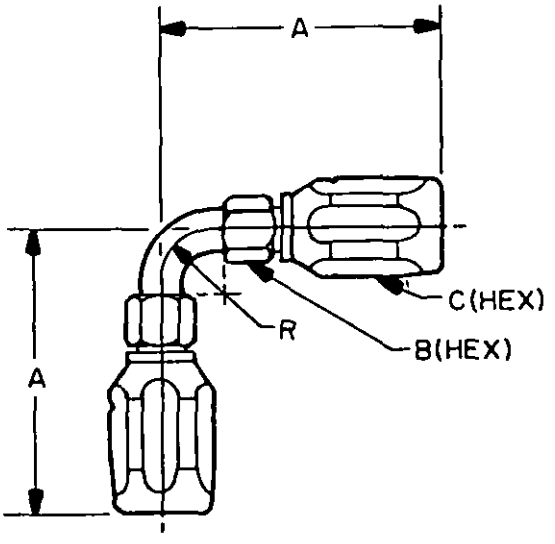
Tube size (nom)	Hose end size (dash no.)	T thread CL-3	A (max)	B (see note 2)	C (see note 2)	H (max)	M (ref)	R rad (min)
1/4	-6	1-13/16-12 UN	3.55	1.25	1.25	2.25	0.62	0.50
3/8	-3	1-3/8-12 UNF	3.81	1.44	1.50	2.50	.62	.75
1/2	-12	1-3/4-12 UN	4.75	1.81	1.75	3.25	.62	1.25
3/4	-16	2-12 UN	5.75	2.12	2.25	3.75	.62	1.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly material for the male half union adapter, hose nipple and socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. The 90-degree tube elbow, for fitting sizes 1/4 through 3/4 inch shall conform to MIL-T-1368, class A.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 5. Type CL, O-ring seal union, male half to hose, 90-degree elbow.

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Tube size (nom)	Rubber hose (dash no.)	A (max)	B (see note 2)	C (see note 2)	R rad (min)
3/8	-6	3.55	0.69	1.25	0.50
1/2	-8	3.81	.90	1.50	.75
3/4	-12	4.75	1.06	1.75	1.25
1	-16	5.75	1.38	2.00	1.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple and sockets shall conform to QQ-N-281, class A or B, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. The 90-degree tube elbow for hose size -6 through -12 shall conform to MIL-T-1368, class B for hose sizes larger than -12 to MIL-T-16420, composition 70-30, grade 1, type I, class 700 special or 1650 special.
4. Socket (body) alternative material shall conform to QQ-N-288, composition B.

FIGURE 6. Type D, 90-degree elbow, hose to hose.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24135/2(SH)	Hose, Synthetic Rubber, Wire Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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